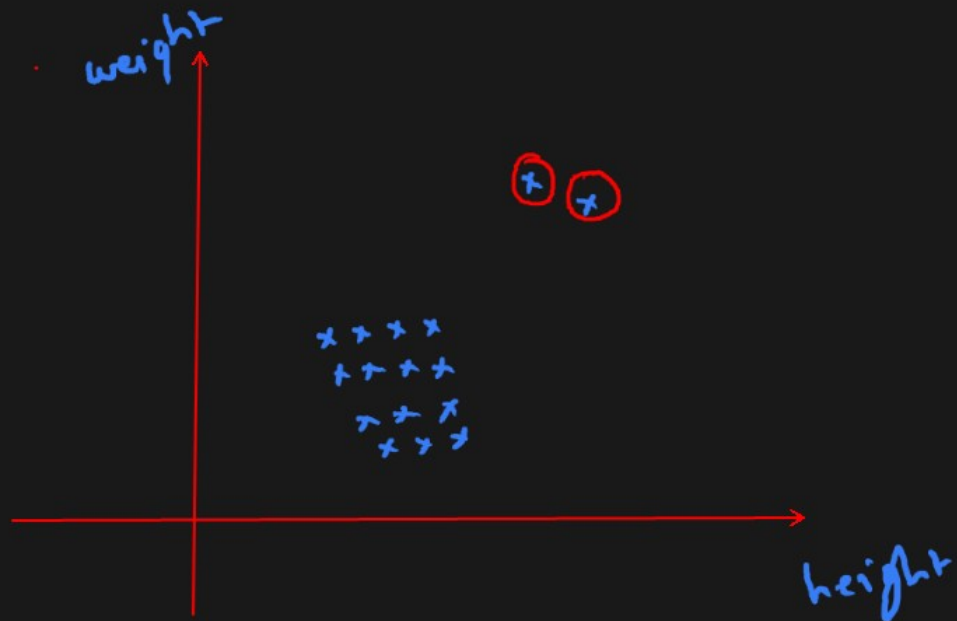
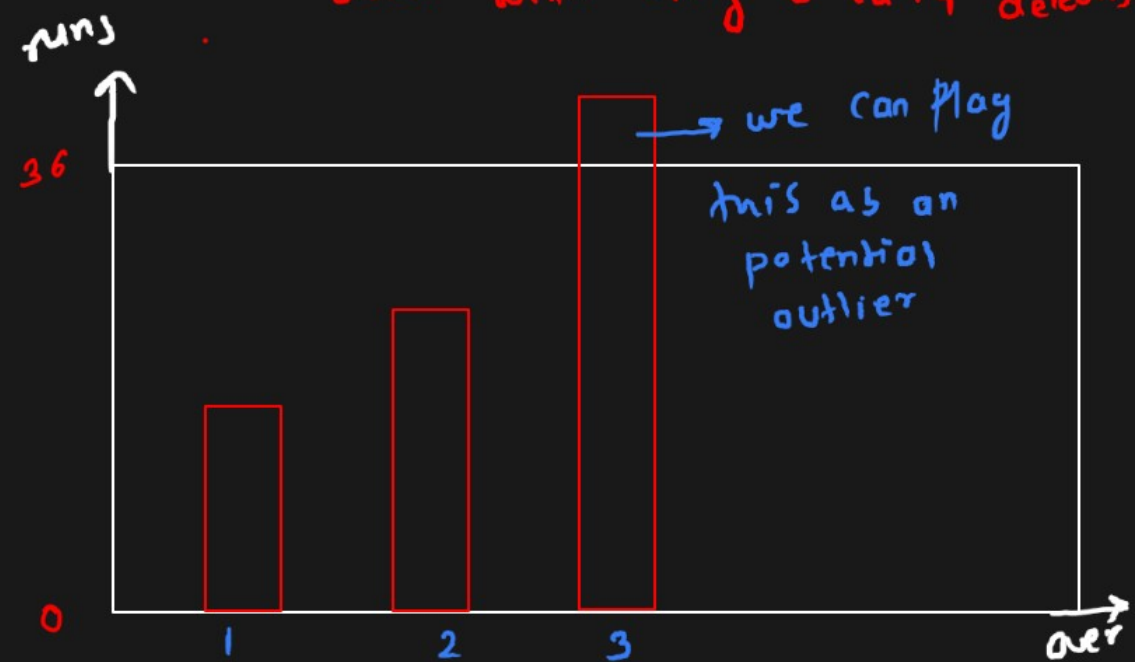


Anomaly Detection : [To detect outliers]



* eg: let's say we are seeing stats of cricket per over and at some point he glitch it happened and it shows 47 runs in an single over with only 6 valid deliveries

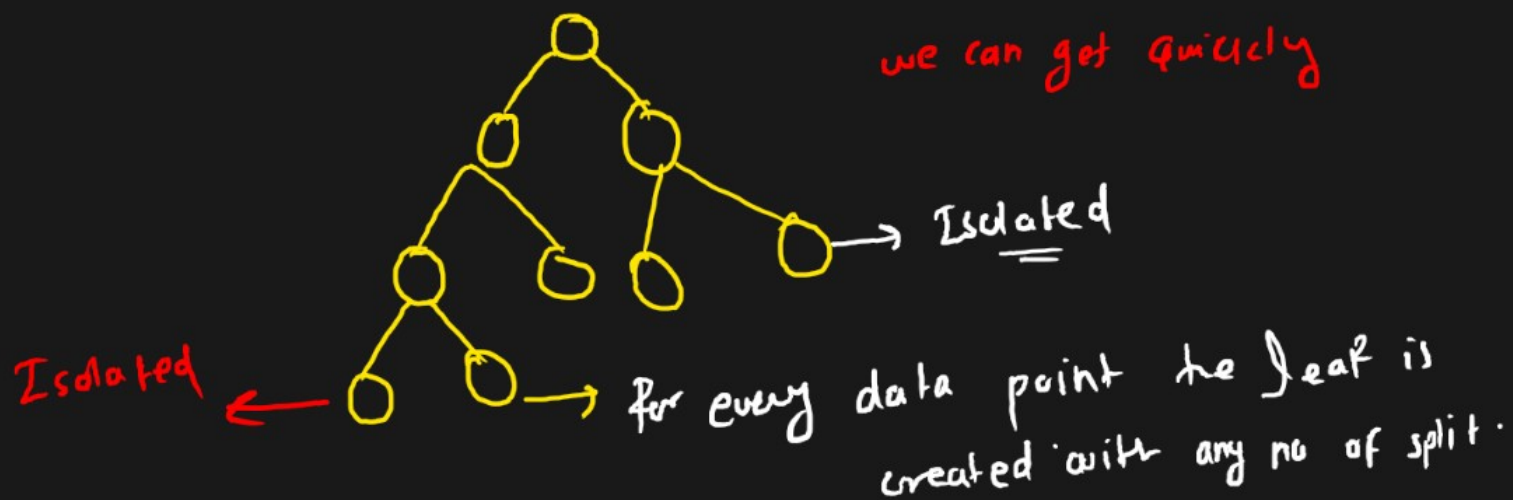
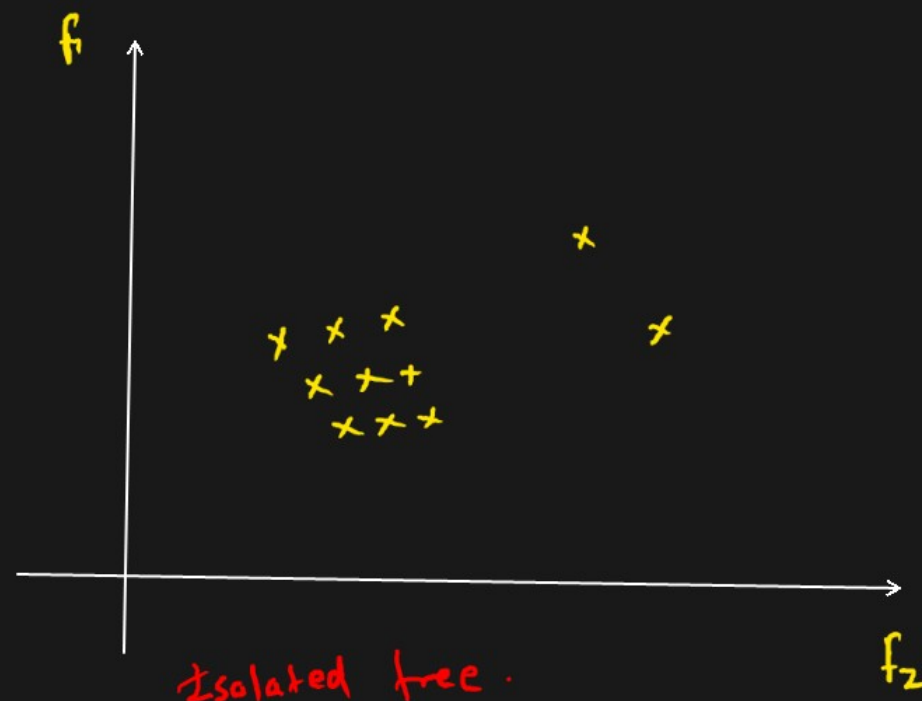


Types :

1] Isolation Forest [Decision Trees]

F_1	F_2	F_3	F_4
-	-	-	-
-	-	-	-
-	-	-	-

→ how quickly we are able to isolate



Anomaly score :-

• mathematical formula to compute anomaly score :-

→ new point

$$S(x|m) = \frac{E(h(x))}{c(m)}$$

$$E(h(x)) < c(m) \Rightarrow S(x|m) \approx 1 \Rightarrow$$

we can set some threshold

anomaly score good

we can consider as outlier

where

m = no of data points

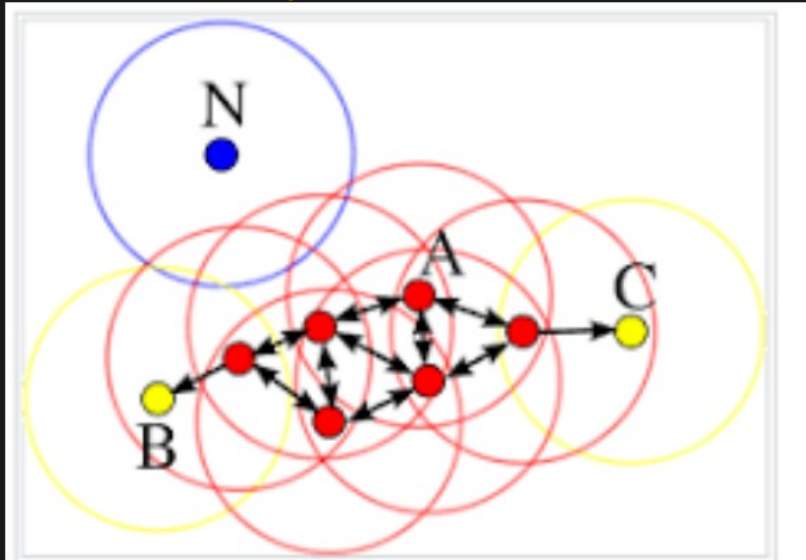
x = data point

$E(h(x))$ = Average search depth for x from the isolation tree for particular point

$c(m)$ = Average value of $h(x)$ for all points

DBSCAN CLUSTERING :-

• Able to cluster in non linear separable data.



• → core points

• → border points

• → noise/outliers

Hyperparameter

① minpts = 4 ② $E = \epsilon$ radius

core points

No of points within the ϵ should



→ core points
coming inside circle

3) Outliers :-



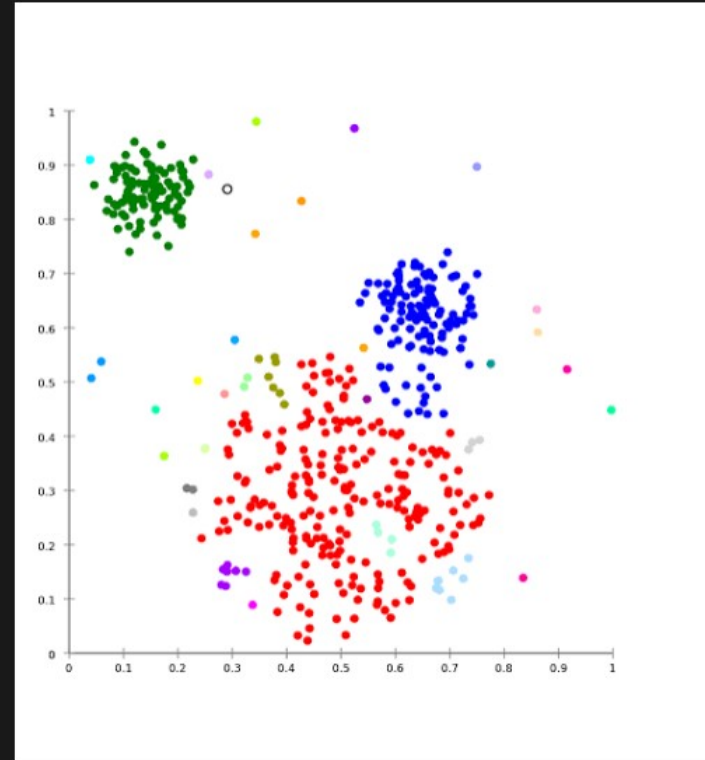
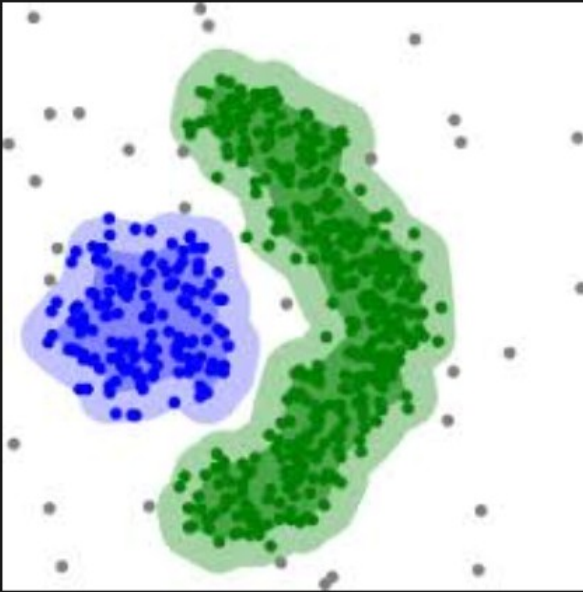
Border points



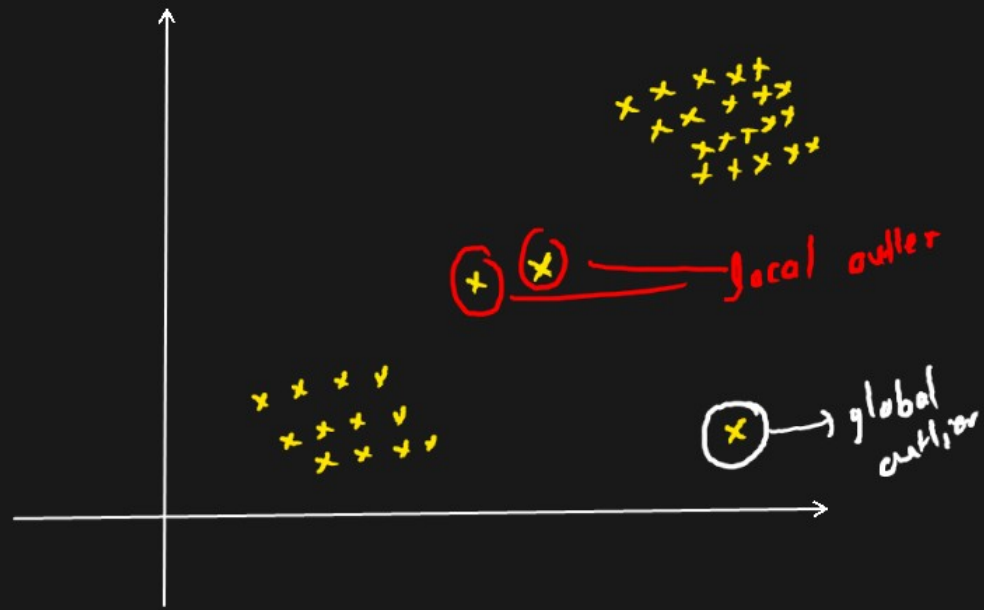
core points < min points

known as border points

Some Examples after applying DB scan clustering



3) Local Outlier Factor :-



→ how well they are isolated

→ use local outlier factor

- 1) local outlier :
- 2) global outlier :

[LOF] $\Rightarrow$ LOF score

internally eval :

k-nearest neighbor concepts.

↓

[local density]